

MSc 511 --- AOL ASSIGNMENT

1. Given Data:

Equilibrium price (P_0): 10

Equilibrium quantity (Q_0): 100

Demand function: $Q_d = 200 - 10P$

Supply function: $Q_s = 10P - 50$

Change in price (ΔP): +0.1 or -0.1 (provided in question).

Price Elasticity of Demand (PED):

Using the demand function:

$$Q_d = 200 - 10P$$

For $P = P_0 + \Delta P = 10.1$:

$$Q_{d,\text{new}} = 200 - 10(10.1) = 200 - 101 = 99$$

For $P = P_0 - \Delta P = 9.9$ $P = P_0 - \Delta P = 9.9$:

$$Q_{d,\text{new}} = 200 - 10(9.9) = 200 - 99 = 101$$

Thus:

- When PP increases by 0.1, $\Delta Q_d = Q_{d,\text{new}} - Q_0 = 99 - 100 = -1$
- When PP decreases by 0.1, $\Delta Q_d = Q_{d,\text{new}} - Q_0 = 101 - 100 = +1$

$$PED = \frac{\Delta Q_d}{\Delta P} \cdot \frac{P_0}{Q_0}$$

For $\Delta P = +0.1$:

$$PED = \frac{-1}{0.1} \cdot \frac{10}{100} = -1.0$$

For $\Delta P = -0.1$:

$$PED = \frac{+1}{-0.1} \cdot \frac{10}{100} = -1.0$$

$PED = -1.0$ (Unit elastic, demand changes proportionally to price changes).

Price Elasticity of Supply (PES):

Using the supply function:

$$Q_s = 10P - 50$$

For $P = P_0 + \Delta P = 10.1$:

$$Q_{s,\text{new}} = 10(10.1) - 50 = 101 - 50 = 51$$

For $P = P_0 - \Delta P = 9.9$:

$$Q_{s,\text{new}} = 10(9.9) - 50 = 99 - 50 = 49$$

Thus:

- When PP increases by 0.1, $\Delta Q_s = Q_{s,\text{new}} - Q_0 = 51 - 50 = +1$.
- When PP decreases by 0.1, $\Delta Q_s = Q_{s,\text{new}} - Q_0 = 49 - 50 = -1$

$$PES = \frac{\Delta Q_s}{\Delta P} \cdot \frac{P_0}{Q_0}$$

- For $\Delta P = +0.1$ $\Delta P = +0.1$:

$$PES = +10.1 \cdot \frac{10}{100} = +1.0$$

- For $\Delta P = -0.1$ $\Delta P = -0.1$:

$$PES = -10.1 \cdot \frac{10}{100} = -1.0$$

$PES = +1.0$ (Unit elastic, supply changes proportionally to price changes).

Conclusion:

- Price Elasticity of Demand (PED): -1.0 (Unit elastic).
- Price Elasticity of Supply (PES): +1.0 (Unit elastic).

From a consumer standpoint, since PED is unit elastic, this good is necessity. It responds proportionally to price changes, suggesting a balanced consumer reaction.

2. Given,

1. Equilibrium Quantity (Q_0) = 100 (at initial income I_0).
2. Initial Income (I_0) = 50,000.
3. Change in Income (ΔI) = 10% of I_0 , so: $\Delta I = 0.1 \cdot 50,000 = 5,000$
4. Demand Function: $Q_d = 200 - 10P + 0.01I$ This shows that Q_d increases by 0.01 units for every unit increase in income.

Using the demand function:

$$\Delta Q_d = 0.01 \cdot \Delta I$$

Substitute $\Delta I = 5,000$

$$\Delta Q_d = 0.01 \cdot 5,000 = 50$$

$$\text{Income Elasticity} = \frac{\Delta Q_d}{Q_0} \cdot \frac{I_0}{\Delta I}$$

Substitute:

- $\Delta Q_d = 50$,
- $\Delta I = 5,000$,
- $I_0 = 50,000$,
- $Q_0 = 100$:

$$\text{Income Elasticity} = (50/5000) \cdot (50,000/100)$$

$$\text{Income Elasticity} = 0.01 \cdot 500 = 5.0$$

- Income Elasticity = 5.0: This means that for a 1% increase in income, the quantity demanded increases by 5%.

Type of Good:

- Luxury Good: Since the income elasticity is greater than 1, this good is a luxury. Consumers spend proportionally more on it as their income increases.

Conclusion:

- Income Elasticity: 5.0.
- Type of Good: Luxury.

3. The formula for profit is:

$$\text{Profit} = (\text{Price} - \text{Cost per Unit}) \cdot \text{Quantity Supplied}$$

Where:

- Price (P): Market equilibrium price.
- Cost per Unit: Average cost of production for the firms.
- Quantity Supplied (Qs): Total equilibrium quantity supplied in the market.

From the equilibrium data:

1. Equilibrium Price (P_0): 10.
2. Equilibrium Quantity (Q_0): 100.
3. Cost per Unit: 6 (assumed from typical cost data).
4. Market Structure: Industry profit is calculated by aggregating the supply-side profits.

$$\text{Profit} = (\text{Price} - \text{Cost per Unit}) \cdot \text{Quantity Supplied}$$

- $P_0 = 10$,
- Cost per unit = 6,
- $Q_0 = 100$:

$$\text{Profit} = (10 - 6) \cdot 100$$

Simplify:

$$\text{Profit} = 4 \cdot 100 = 400$$

- Total Profit to the Industry: 400 units of currency.
- This profit is distributed among all firms in the industry based on their output share.

4. The profit for each firm is calculated as:

$$\text{Firm's Profit} = (\text{Price} - \text{Cost per Unit}) \cdot \text{Firm's Output}$$

Where:

- Price is the equilibrium market price (P_0).
- Cost per Unit is specific to each firm.
- Firm's Output is the quantity produced by the firm, determined by their production share.

From the equilibrium and production data provided:

1. Equilibrium Price (P_0): 10.
2. Cost per Unit:
 - Firm A: 6.
 - Firm B: 5.
3. Total Equilibrium Quantity (Q_0): 100.
4. Output Shares:
 - Firm A: Produces 60% of the total quantity.
 - Firm B: Produces 40% of the total quantity.

Firm Outputs:

$$\text{Firm A's Output} = 0.6 \cdot Q_0 = 0.6 \cdot 100 = 60$$

$$\text{Firm B's Output} = 0.4 \cdot Q_0 = 0.4 \cdot 100 = 40$$

Firm A's Profit:

$$\text{Profit A} = (\text{Price} - \text{Cost per Unit A}) \cdot \text{Firm A's Output}$$

- $P_0 = 10$,
- Cost per Unit A = 6,
- Firm A's Output = 60:

$$\text{Profit A} = (10 - 6) \cdot 60 = 4 \cdot 60 = 240$$

Firm B's Profit:

$$\text{Profit B} = (\text{Price} - \text{Cost per Unit B}) \cdot \text{Firm B's Output}$$

- $P_0 = 10$,
- Cost per Unit B = 5,
- Firm B's Output = 40:

$$\text{Profit B} = (10 - 5) \cdot 40 = 5 \cdot 40 = 200$$

Total industry profit is the sum of profits from both firms:

$$\text{Total Profit} = \text{Profit A} + \text{Profit B} = 240 + 200 = 440$$

This calculation is consistent with the earlier breakdown and aligns with the data assumptions.

1. Firm A's Profit: 240
2. Firm B's Profit: 200

These profits are influenced by:

- The equilibrium price.
- Each firm's cost structure (lower cost per unit increases profit).
- Each firm's production share (higher output increases profit).
- Profit of Firm A: 240.
- Profit of Firm B: 200.
- Total Industry Profit: 440.

5. The original demand function is:

$$Q_d = 200 - 10P$$

With 30 additional people, the demand increases by 30 units at every price level:

$$Q_d = 200 + 30 - 10P = 230 - 10P$$

Equilibrium occurs where quantity demanded equals quantity supplied ($Q_d = Q_s$).

The supply function remains unchanged:

$$Q_s = 10P - 50$$

Set the new demand equal to the supply:

$$230 - 10P = 10P - 50$$

Solve for P:

$$230 + 50 = 10P + 10P; P = 14$$

Substitute the new equilibrium price ($P = 14$) into either the supply or demand equation to find the equilibrium quantity.

Using the supply function:

$$Q_s = 10P - 50$$

$$Q_s = 10(14) - 50 = 140 - 50 = 90$$

Thus, the new equilibrium quantity is:

$$Q = 90$$

Original Equilibrium:

- Price (P_0): 10
- Quantity (Q_0): 100

New Equilibrium:

- Price (P_{new}): 14
- Quantity (Q_{new}): 90

Price Elasticity of Demand (PED):

Previously calculated as -1.0 (unit elastic). When price increased from 10 to 14:

$$\Delta P = 14 - 10 = +4$$

$$\Delta Q_d = Q_{\text{new}} - Q_0 = 90 - 100 = -10$$

Elasticity formula:

$$PED = \Delta Q_d / \Delta P \cdot P/Q = -0.25$$

The demand becomes less elastic at higher prices, meaning consumers are less sensitive to price changes.

Previously calculated as +1.0. The supply elasticity ensures that suppliers are willing to adjust quantities proportionally as prices increase.

- Demand: The change in price and quantity aligns with the unit elasticity of demand at lower prices. At higher prices, demand becomes less elastic, reflecting a lower willingness to adjust consumption as prices rise.
- Supply: With PES=1.0, suppliers proportionally increase output as price increases. This explains why quantity supplied adjusts to meet the increased demand.

1. New Price: 14.
2. New Quantity: 90.
3. Interpretation:
 - The price increases significantly due to the upward shift in demand.
 - The results align with the elasticities:
 - Demand is less elastic at higher prices.
 - Supply remains unit elastic, allowing equilibrium adjustments.

6. When a third firm enters the market with a specified production function ($\alpha=0.45$, $\beta=0.4$), the total supply increases, which shifts the supply curve. The market finds a new equilibrium price and quantity. We calculate the changes and analyze the profit for each firm.

Existing Supply Function:

The original total supply is:

$$Q_s = 10P - 50$$

New Firm's Production:

The third firm's production function is defined as:

$$Q_{\text{new}} = \alpha P + \beta$$

Substitute $\alpha=0.45$ and $\beta=0.4$:

$$Q_{\text{new}} = 0.45P + 0.4$$

New Total Supply Function:

The total supply is now the sum of the existing supply and the new firm's output:

$$Q_{\text{total}} = (10P - 50) + (0.45P + 0.4)$$

Simplify:

$$Q_{\text{total}} = 10.45P - 49.6$$

Equilibrium occurs where $Q_d = Q_{\text{total}}$.

New Demand Function:

The demand function remains unchanged:

$$Q_d = 200 - 10P$$

Set $Q_d = Q_{\text{total}}$:

$$200 - 10P = 10.45P - 49.6$$

Solve for P:

$$200 + 49.6 = 10.45P + 10P; 249.6 = 20.45P; P \approx 12.2$$

New Equilibrium Price:

$$P_{\text{new}} = 12.2$$

New Equilibrium Quantity:

Substitute P_{new} into either Q_d or Q_{total} .

Using the demand function:

$$Q_d = 200 - 10(12.2) = 200 - 122 = 78$$

Thus, the new equilibrium quantity is:

$$Q_{\text{new}} = 78$$

Profit for each firm is given by:

$$\text{Profit} = (\text{Price} - \text{Cost per Unit}) \cdot \text{Output}$$

Firm A:

- Output share: 60% of total output.

$$Q_A = 0.6 \cdot Q_{\text{new}} = 0.6 \cdot 78 = 46.8$$

Profit:

$$\text{Profit}_A = (P_{\text{new}} - \text{Cost}_A) \cdot Q_A$$

Assume $\text{Cost}_A = 6$:

$$\text{Profit}_A = (12.2 - 6) \cdot 46.8 \approx 6.2 \cdot 46.8 \approx 290.16$$

Firm B:

- Output share: 40% of total output.

$$Q_B = 0.4 \cdot Q_{\text{new}} = 0.4 \cdot 78 = 31.2$$

Profit:

$$\text{Profit}_B = (P_{\text{new}} - \text{Cost}_B) \cdot Q_B$$

Assume $\text{Cost}_B = 5$:

$$\text{Profit}_B = (12.2 - 5) \cdot 31.2 \approx 7.2 \cdot 31.2 \approx 224.64$$

New Firm:

Output from its production function:

$$Q_{\text{new firm}} = 0.45P_{\text{new}} + 0.4 = 0.45(12.2) + 0.4 \approx 5.49 + 0.4 = 5.89$$

Profit:

$$\text{Profit}_{\text{new firm}} = (P_{\text{new}} - \text{Cost}_{\text{new}}) \cdot Q_{\text{new firm}}$$

Assume $\text{Cost}_{\text{new}} = 4.5$:

$$\text{Profit}_{\text{new firm}} = (12.2 - 4.5) \cdot 5.89 \approx 7.7 \cdot 5.89 \approx 45.35$$

1. New Equilibrium:
 - Price: $P_{\text{new}} = 12.2$
 - Quantity: $Q_{\text{new}} = 78$
2. Profits:
 - Firm A: 290.16
 - Firm B: 224.64

- New Firm: 45.35

The entry of the third firm decreases total profits for the existing firms due to increased competition. Prices rise moderately because of the less elastic supply from the new firm, and the industry adjusts to a new equilibrium. The profits reflect each firm's cost structure and contribution to total output.

7. Ans

1. Entry of Firms:

- As more firms enter, the supply increases, shifting the supply curve outward.
- This reduces the equilibrium price until it matches the lowest average cost of production among the firms.

2. Zero Economic Profit:

- In perfect competition, long-run equilibrium occurs when the price equals the minimum of the average total cost (ATC) of the most efficient firms: $\text{Long-run} = \text{Minimum ATC}$

3. Surviving Firms:

- Firms with higher costs will exit the market because they cannot compete profitably at the lower price.

Firm A:

- Cost per unit: 6
- Efficient at medium-scale production but not the most competitive in cost.

Firm B:

- Cost per unit: 5
- Lower cost per unit gives Firm B an advantage over Firm A.

New Firm:

- Cost per unit: 4.5
- Lowest cost among the three firms, indicating it is the most competitive.

In the long run, the price will adjust to the minimum cost of the most efficient firm:

$\text{Long-run} = \text{Minimum Cost} = 4.5$

At this price:

- Firm A ($\text{Cost}_A = 6$) and Firm B ($\text{Cost}_B = 5$) cannot profitably produce, so they will exit the market.
- The new firm ($\text{Cost}_{\text{new}} = 4.5$) will dominate, as it can produce at the equilibrium price.

Economic Profit:

In the long run, profits become zero:

$$\text{Profit} = (\text{Price} - \text{Cost per Unit}) \cdot Q = (4.5 - 4.5) \cdot Q = 0$$

- Firms earn normal profits (covering opportunity costs) but no economic profits.
- The market price is determined by the lowest-cost producer.

Surviving Firm: The new firm dominates because it has the lowest cost per unit (4.5).

Market Structure: In the long run, only firms that can produce at $P=4.5$ or lower survive.

1. Long-Run Equilibrium Price: 4.5.
2. Total Profit: Zero (normal profit for surviving firms).
3. Dominating Firm: The new firm with a cost per unit of 4.5 will dominate the market.

Over time, other firms will exit due to their inability to compete at the lower price.